

Sequences

Recurrence sequences are sequences that are defined inductively: apart from the initial terms, each consequent term of the sequence is given as a function of previous terms and the index.

A very important class of recurrence sequences for which there is a method of solving are the *linear recurrence sequences*. These sequences satisfy a linear relation of the form

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k}, \quad (*)$$

where $c_1, c_2, \dots, c_k$ are constants. The first step in solving such sequences is finding zeros of its *characteristic polynomial*:

$$P(x) = x^k - c_1 x^{k-1} - c_2 x^{k-2} - \cdots - c_k.$$

Indeed, if ξ is its zero of multiplicity r , then each of the r sequences $\xi^n, n\xi^n, \dots, n^{r-1}\xi^n$ satisfies the recurrence relation (*). Since there are k zeros counted with multiplicities, this yields exactly k sequences that satisfy (*). In fact, every sequence satisfying (*) is a linear combination of these k sequences:

Theorem. (Linear recurrences)

Let the characteristic polynomial of (*) be $P(x) = (x - \alpha_1)^{r_1} (x - \alpha_2)^{r_2} \cdots (x - \alpha_\ell)^{r_\ell}$, where $r_1 + \cdots + r_\ell = k$. Then

$$a_n = Q_1(n)\alpha_1^n + Q_2(n)\alpha_2^n + \cdots + Q_\ell(n)\alpha_\ell^n,$$

where $Q_i(x)$ are some polynomials with $\deg Q_i < r_i$. The polynomials Q_i are obtained from the k initial terms, by solving a system of linear equations.

It comes without saying that not every recurrence can be solved in a closed form. Sometimes all we have is the recurrence.



- 1 Define a sequence (a_n) by $a_0 = 0$, $a_1 = 1$ and $a_{n+1} = \frac{8}{5}a_n - a_{n-1}$ for $n \geq 1$. Prove that $|a_n| \leq \frac{5}{3}$ for all n .
- 2 Define $a_0 = 1$, $a_1 = 2$ and $a_{n+1} = 4a_n - a_{n-1}$ for $n \geq 1$. Prove that $a_{2n} = 2a_n^2 - 1$ for all n .
- 3 Given constants A and B , define $a_0 = A$, $a_1 = B$ and $a_{n+1} = \frac{1}{a_n} - a_{n-1}$ for $n \geq 1$. Find a_n .
- 4 Given a real number a , we define a sequence by $x_0 = 1$, $x_1 = x_2 = a$, and $x_{n+1} = 2x_n x_{n-1} - x_{n-2}$ for $n \geq 2$. Prove that if $x_m = 0$ for some m , then the sequence is periodic.
- 5 Define $a_0 = 0$, $a_1 = 1$ and $a_{n+1} = 2a_n + a_{n-1}$. Prove that $2^k \mid a_n$ if and only if $2^k \mid n$.
- 6 Prove that all terms of the sequence (a_n) defined by $a_0 = 1$ and $a_{n+1} = 3a_n + \sqrt{8a_n^2 + 1}$ for $n \geq 0$ are integers.
- 7 A sequence (y_n) is defined by $y_1 = y_2 = y_3 = 1$ and $y_{n+1} = \frac{y_n y_{n-1} + 1}{y_{n-2}}$ for $n \geq 3$. Prove that all terms of this sequence are integers.
- 8 Find all sequences of positive integers (a_n) such that $a_n + a_{n+1} = a_{n+2} a_{n+3} - 1001$ for all n .
- 9 A sequence (a_n) is defined by $a_1 = a_2 = a_3 = a_4 = 1$ and $a_n a_{n+4} = a_{n+1} a_{n+3} + a_{n+2}^2$ for $n \geq 1$. Prove that each a_n is an integer. (This is called the *Somos-4 sequence*.)
- 10 Define $a * b = \frac{a+b}{ab+1}$. Evaluate $((((2 * 3) * 4) * 5) * \cdots * 100)$.
- 11 Let $a_1 = 1$ and $a_1 + a_2 + \cdots + a_n = n^2 a_n$ for all $n \geq 1$. Compute a_n .

- 12 Define $a_1 = 1$, $b_1 = 2$ and $a_{n+1} = \frac{1+a_n+a_nb_n}{b_n}$, $b_{n+1} = \frac{1+b_n+a_nb_n}{a_n}$. Prove that $a_{1000} < 5$.
- 13 Define $a_0 = 2222$ and $a_{n+1} = \frac{a_n^2}{1+a_n}$ for $n = 1, 2, \dots$. Evaluate $\lfloor a_{1000} \rfloor$.
- 14 Define $x_1 = \frac{1}{2}$ and $x_{n+1} = 1 - x_1x_2 \cdots x_n$ for $n \geq 1$. Prove that $x_{100} > 0.99$.
- 15 The sequence (a_n) is defined by $a_1 = \frac{1}{2}$ and $a_n = \frac{2n-3}{2n}a_{n-1}$ for $n > 1$.
(a) Prove that $a_1 + a_2 + \cdots + a_n < 1$ for all $n \geq 1$.
(b) Prove that $a_1 + a_2 + \cdots + a_n > 0.999$ if n is large enough.
- 16 If $x_1, x_2, \dots$ are positive and $\sum_{n=1}^{\infty} \frac{x_n}{2^{n-1}} = 1$, prove that $\sum_{n=1}^{\infty} \frac{x_1 + \cdots + x_n}{n^2} \leq 2$.
- 17 A sequence of nonnegative integers $a_0 = 1, a_1, a_2, \dots$ satisfies the inequality $a_n \geq a_{n+1} + a_{n+2}$ for every $n \geq 0$. Find the smallest constant C such that $a_1 + a_2 + \cdots \leq C$ holds for every such sequence.
- 18 Suppose that $a > 0$ and real numbers $x_1, x_2, \dots, x_n$ with $n \geq 4$ are such that $x_i^2 = x_{i+1}x_{i+2} + a$ for $i = 1, \dots, n$ (the indices are mod n). Prove that at least two of the numbers $x_1, \dots, x_n$ are negative.
- 19 Find all integers $n \geq 3$ for which there exist real numbers $a_1, a_2, \dots, a_n$ satisfying $a_{i+2} = a_i a_{i+1} + 1$ for $i = 1, 2, \dots, n$. (The indices are counted modulo n .)
- 20 A sequence of positive numbers (a_i) satisfies $a_{i+j} \leq a_i + a_j$ for all $i, j \in \mathbb{N}$.
(a) Prove that $a_1 + a_2 + \cdots + a_k \geq \frac{k(k+1)}{2m} a_m$ for any integers $m \geq k > 0$.
(b) Prove that $\frac{a_1}{1} + \frac{a_2}{2} + \cdots + \frac{a_n}{n} \geq n \cdot \frac{a_m}{m}$ for any positive integers $m \geq n$.
- 21 An infinite sequence $a_1, a_2, a_3, \dots$ of nonnegative real numbers satisfies $a_n - 2a_{n+1} + a_{n+2} \geq 0$ and $a_1 + a_2 + \cdots + a_n \leq 1$ for all n . Prove that $0 \leq a_n - a_{n+1} < \frac{2}{n^2}$.
- 22 Positive integers $b_1, b_2, \dots$ satisfy $1 = \frac{b_1}{1^2} > \frac{b_2}{2^2} > \frac{b_3}{3^2} > \dots$. If r is the largest real number satisfying $\frac{b_n}{n^2} \geq r$ for all n , what are the possible values of r ?
- 23 A sequence $a_1, a_2, \dots$ of nonnegative integers satisfies $a_{m+n} - a_m - a_n \in \{0, 1\}$ for all m, n . Prove that there is a real number α such that $a_n = \lfloor n\alpha \rfloor$ for every n .
- 24 Find all sequences of positive integers $a_1, a_2, \dots$ such that $\lfloor a_m/a_n \rfloor = \lfloor m/n \rfloor$ holds for all m and n .
- 25 Real numbers $x_1, \dots, x_n$ are such that among the values $\frac{x_1 + \cdots + x_k}{k}$ for $k = 1, \dots, n$ the smallest is 0 and the largest is 1. Find the largest and smallest possible values of $\max_{i,j} (a_i - a_j)$.
- 26 Does there exist a bounded infinite sequence $a_1, a_2, a_3, \dots$ with the property that $|a_i - a_j| \geq \frac{1}{|i-j|}$ for any two positive integers $i \neq j$?
- 27 Let $c > 0$ be a real number. We want to find an infinite sequence $a_1, a_2, a_3, \dots \in [0, c]$ that satisfies $|a_i - a_j| > \frac{1}{i+j}$ for any positive integers $i \neq j$. Prove that:
(a) there is no such sequence if $c < 1$; (b) such a sequence exists if $c = 1.4$.
- 28 A sequence of real numbers $a_1, a_2, \dots$ satisfies $a_n = -\max_{i+j=n} (a_i + a_j)$ for all $n > 1000$. Prove that the sequence is bounded.
- 29 Real numbers $x_1, \dots, x_n$ with the zero-sum are not all zero. Suppose c is a constant such that $x_i \leq x_{i+1} + c \max\{0, x_{i+2}\}$. Prove that $c \geq 2$, and that $c = 2$ is possible if and only if n is even.